

SUPPORT, NOT A CLAIM OF RESULTS

Shape $I(t)$ at the issue queue so less decoupling C is needed

Pack high-current uops, then low-current uops. Control the excitation. Do not lock the story to a buck converter.

Sources: our deck *Powering Superscalar Processors*; Lin & Chang, EUC 2006 (LinC06.pdf). BOOM fields from Chipyard 1.3 IssueUnitCollapsing.

PAPERS ALREADY ON THE TABLE

PAPER	WHAT IT ACTUALLY DOES	WHAT IT DOES NOT
Kumar, Alsalam, Amrouch, Gerstlauer, <i>IEEE TC</i> 2023 (LACP _o)	Activity → decision trees → cycle $P(t)$ on BOOM. Oracle for $I(t)=P/V$.	Does not schedule. Does not size decap.
Meng & Saleh, <i>IEEE JSSC</i> 2009	Active decap. Senses V , switches C . 10–20% noise vs passive, 90 nm.	Reacts to droop. No instruction stream. No lead time.
Lin & Chang, EUC 2006 (LinC06)	Compiler: merge FU-alike regions, DVS, turn idle ALUs/MULs off. Wattch. ~26% energy, ~12–18% slowdown.	Not hardware issue select. Not PDN. Not current class. Not C .
Pathania, HotSniper (2021) — in our deck	Thermal interval sim. Supports “hotspot ≠ whole-core throttle.”	Not PDN, not issue queue.

HONEST GAP

LinC06 groups work that uses the same FU so that unit can sleep longer. Energy via DVS / FU gating.

Meng dumps charge after V already moved.

LACP0 estimates $P(t)$ after the fact.

Nobody in these refs selects ready uops by predicted current, at the IQ grant loop, to cut on-die C.

Closest compiler cousin: Rele et al. (cited in LinC06) — idle FU gating on a superscalar. Still not PDN, still not grant.

MEDIUM BOOM — WHAT ACTUALLY PICKS THE NEXT UOP

Age-ordered collapsing IQ. Oldest = slot 0. No age tag.

WIRE / FIELD	JOB
p1 p2 p3 ppred	Operands present → request
fu_code [9:0]	Port legal: (fu_code & fu_types) ≠ 0
uopc [8:0]	Opcode class (ALU / mul / FMA / ...)
state	Empty / one uop / split store
slot index	Priority. First request that fits a free port gets grant
is_br / is_jal / is_jalr	request_hp only in unordered IQ. Medium BOOM does not use this path.
No current bit exists. LACPo logs state, uopc, fu_code per slot — not request / grant / p*.	

PROPOSED CHANGE

In the grant loop: ready $\wedge$ FU-legal $\wedge$ **current class**.

- Class from uopc / fu_code, or one extra MicroOp bit.
- High window, then low window. Same ILP budget, different I(t) envelope.
- Goal: $C_{\text{needed}} \downarrow$ because di/dt is scheduled, not only filtered.

Active dump / tokens (our existing deck) stay optional for tails. IQ shaping is the new claim.

QUESTION FOR MA'AM

This is instruction-side current shaping. Converter topology is not the claim.

- Where does her controller live — board VRM, package, on-die IVR?
- Is a 2-level current hint from the IQ a useful input to that loop?
- What latency can that loop actually use? (~30 clk token vs 0–2 clk issue fire)

WHAT IS TRUE TODAY

- LACPo path works on dummy VCD. Dummy I(t) is not RTL.
- Lead-time numbers in our deck are schematic, not a measured VCD study.
- MediumBoom debug sim is built. load_step smoke is running. Real VCD is next.

Kumar et al., IEEE TC 2023, doi:10.1109/TC.2022.3185572 · Meng & Saleh, IEEE JSSC 2009, doi:10.1109/JSSC.2008.2010752 · Lin & Chang, EUC 2006, doi:10.1007/11802167_6